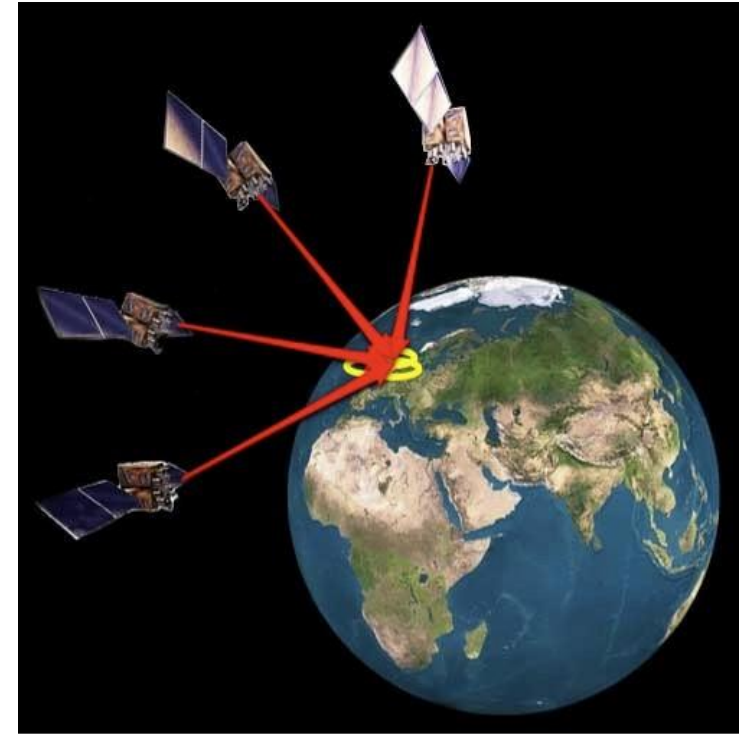


Source Geolocation using Frequency Measurements

ENEL445 - 26S1 Project

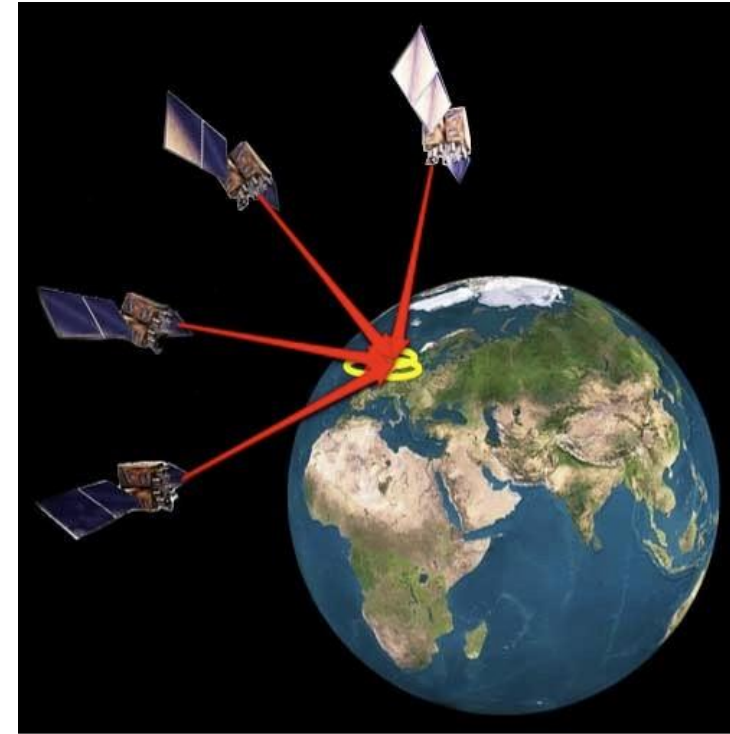
Problem Description (1)

- **Source geolocation** refers to estimating the geodesic location of a signal emitter on the Earth surface from signal measurements
- A large variety of **civil and military applications**
 - Navigation, search and rescue, surveillance, jammer localization ...
- Source geolocation using frequency measurements
 - Time measurements such as ranging are inaccurate for narrow-band sources



Problem Description (2)

- Source geolocation using **frequency difference of arrival (FDOA)** measurements **from four satellites**
- The satellites are moving, causing relative motion between the satellites and the ground source
- The source signal received at each satellite is then subject to Doppler shift, which depends on the source location
- The source signal frequency is unknown, necessitating using FDOA rather than frequency measurements for geolocation



Mathematical Preliminary

- Source position in Earth Centered Earth Fixed coordinate system:

$$\mathbf{u}^o = [x^o, y^o, z^o]^T$$

- Source geodetic position (latitude, longitude, altitude):

$$\mathbf{p}^o = [B^o, L^o, H^o]^T$$

- Coordinate conversion:

$$x^o = (R_E(B^o) + H)\cos(B^o)\cos(L^o),$$

$$y^o = (R_E(B^o) + H)\cos(B^o)\sin(L^o),$$

$$z^o = [(1 - e^2)R_E(B^o) + H]\sin(B^o),$$

$$R_E(B^o) = \frac{R_0}{\sqrt{1 - e^2\sin^2(B^o)}},$$

$$e = 0.081819198425 \quad R_0 = 6378.137\text{km}$$

Doppler and FDOA

- Satellite ECEF positions and velocities:

$$\mathbf{s}_i = [s_{x,i}, s_{y,i}, s_{z,i}]^T \text{ and } \dot{\mathbf{s}}_i = [\dot{s}_{x,i}, \dot{s}_{y,i}, \dot{s}_{z,i}]^T \quad i = 1, 2, 3, 4$$

- Received frequency at satellite i :

$$f_i = f_0 + \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_i\|} + n_i$$

- FDOA measurements ($j = 2, 3, 4$):

$$f_{j1} = f_j - f_1 = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_j)^T \dot{\mathbf{s}}_j}{\|\mathbf{u}^o - \mathbf{s}_j\|} - \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_1)^T \dot{\mathbf{s}}_1}{\|\mathbf{u}^o - \mathbf{s}_1\|} + (n_j - n_1)$$

Vector Form Representation

Measured FDOAs

$$\mathbf{f} = [f_{21}, f_{31}, f_{41}]^T$$

True FDOAs

$$\mathbf{g}(B^o, L^o) = \begin{bmatrix} g(B^o, L^o, \mathbf{s}_2, \dot{\mathbf{s}}_2) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_3, \dot{\mathbf{s}}_3) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_4, \dot{\mathbf{s}}_4) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \end{bmatrix}$$

Noise

$$\mathbf{n}_f = \begin{bmatrix} n_2 - n_1 \\ n_3 - n_1 \\ n_4 - n_1 \end{bmatrix}$$

$$g(B^o, L^o, \mathbf{s}_i, \dot{\mathbf{s}}_i) = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_j\|}$$

Phase-1 of the Project (1)

- Assume the noise in the frequency measurements n_i is Gaussian with zero mean and variance σ_f^2

$$f_i = f_0 + \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_i\|} + n_i$$

- The FDOA measurements are subject to zero-mean Gaussian noise too

$$f_{j1} = f_j - f_1 = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_j)^T \dot{\mathbf{s}}_j}{\|\mathbf{u}^o - \mathbf{s}_j\|} - \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_1)^T \dot{\mathbf{s}}_1}{\|\mathbf{u}^o - \mathbf{s}_1\|} + (n_j - n_1)$$

- Derive the covariance of the FDOA noise vector $\mathbf{n}_f = \begin{bmatrix} n_2 - n_1 \\ n_3 - n_1 \\ n_4 - n_1 \end{bmatrix}$

Phase-1 of the Project (2)

- **Formulate** the source geolocation problem as a maximum-likelihood estimation problem (see reference [1]) and **transforms** it into a non-linear least-square (NLS) problem

- Grid search-based solution

$$\begin{array}{ll} f_0 = 1\text{GHz} & \mathbf{s}_1 = [7378.1, \quad 0, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_1 = [\quad 0.0001, 4.4995, 5.3623]^T \text{ km/s}, \\ \mathbf{p}^o = [5^\circ, 10^\circ, 0]^T & \mathbf{s}_2 = [7377.5, \quad 100, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_2 = [-0.0671, 4.9493, 4.9497]^T \text{ km/s}, \\ & \mathbf{s}_3 = [7377.5, -100, \quad 0]^T \text{ km}, \dot{\mathbf{s}}_3 = [\quad 0.0610, 4.4991, 5.3623]^T \text{ km/s}, \\ & \mathbf{s}_4 = [7377.5, \quad 0, \quad 100]^T \text{ km}, \dot{\mathbf{s}}_4 = [-0.0777, 4.0150, 5.7335]^T \text{ km/s}. \end{array}$$

- Perform a grid search over $[0, 40^\circ] \times [0, 40^\circ]$, output the one with the smallest cost function value

Phase-1 of the Project (3)

- Monte Carlo simulation-based performance evaluation
 - For each setup, perform $L = 10,000$ ensemble run
 - In each run, generate new random Gaussian noise and add to the true FDOA
 - Perform the grid search-based method to find the geolocation result
 - Compute the estimation root mean square error (RMSE) using

$$\text{RMSE} = \sqrt{\frac{1}{L} \sum_{l=1}^L \|\mathbf{u}_l - \mathbf{u}^o\|^2},$$

- Consider four scenarios, each with different frequency noise levels ($\sigma_f = 1, 2, 4, 8, 16\text{Hz}$)
- Plot the results

Phase-2 of the Project (1)

- Solve the non-linear least square problem formulated in Phase-1 using a gradient-based approach
- Q1: With FDOA, the knowledge on the signal wavelength is still required:

$$f_{j1} = f_j - f_1 = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_j)^T \dot{\mathbf{s}}_j}{\|\mathbf{u}^o - \mathbf{s}_j\|} - \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_1)^T \dot{\mathbf{s}}_1}{\|\mathbf{u}^o - \mathbf{s}_1\|} + (n_j - n_1)$$

- Find a way to estimate it and justify your method through an accuracy analysis

Phase-2 of the Project (2)

- Bonus Question (5%): we use three FDOAs in this project $\mathbf{f} = [f_{21}, f_{31}, f_{41}]^T$
- Can the use of e.g., f_{23} , f_{43} etc together with $\mathbf{f} = [f_{21}, f_{31}, f_{41}]^T$ improve performance?
- Q2: Derive analytically the Jacobian of the FDOA measurement function using the chain rule

$$\mathbf{g}(B^o, L^o) = \begin{bmatrix} g(B^o, L^o, \mathbf{s}_2, \dot{\mathbf{s}}_2) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_3, \dot{\mathbf{s}}_3) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \\ g(B^o, L^o, \mathbf{s}_4, \dot{\mathbf{s}}_4) - g(B^o, L^o, \mathbf{s}_1, \dot{\mathbf{s}}_1) \end{bmatrix} \quad g(B^o, L^o, \mathbf{s}_i, \dot{\mathbf{s}}_i) = \frac{1}{\lambda} \cdot \frac{(\mathbf{u}^o - \mathbf{s}_i)^T \dot{\mathbf{s}}_i}{\|\mathbf{u}^o - \mathbf{s}_j\|}$$

Phase-2 of the Project (3)

- Q3: Verify your analytical results using the finite difference method for computing the gradient/derivatives numerically
- Bonus Question (10%): verify your gradient results using the complex step method
- Q4: Develop (design + implement + test) a gradient-based approach for source geolocation
 - Conjugate gradient, Newton's method, Gauss Newton method, Levenberg-Marquardt, BFGS ...
 - With line search, stopping criteria, multi-start strategy
 - Make sure your iterations always stay in the area of interest $[0, 40^\circ] \times [0, 40^\circ]$

Phase-2 of the Project (4)

- Q5: Compare the RMSE performance of your gradient-based approach with the grid search-based solution in Phase-1 using Monte Carlo simulation
- Bonus question (15%): implement a gradient-free technique such as the particle swarm optimization (PSO) algorithm for source geolocation and compare its performance with the gradient-based method and grid search-based solution.

References

- [1]. [Source geolocation project description](#)
- [2]. Y. Pei et al. “A closed-form solution for source localization using FDOA measurements only,” *IEEE Communications Letters*, 2022.